

Name: 	Labor Protokoll	Datum: 23. Sep. 2025
Klasse: 7A CIF Gruppe: 2		Übung Nr.: LÜ-03 Raum Nr.: B3.11
Lehrer:	Abgabedatum: 25. Mai. 2026	Bewertung:

RIP Kreistopologie LAB

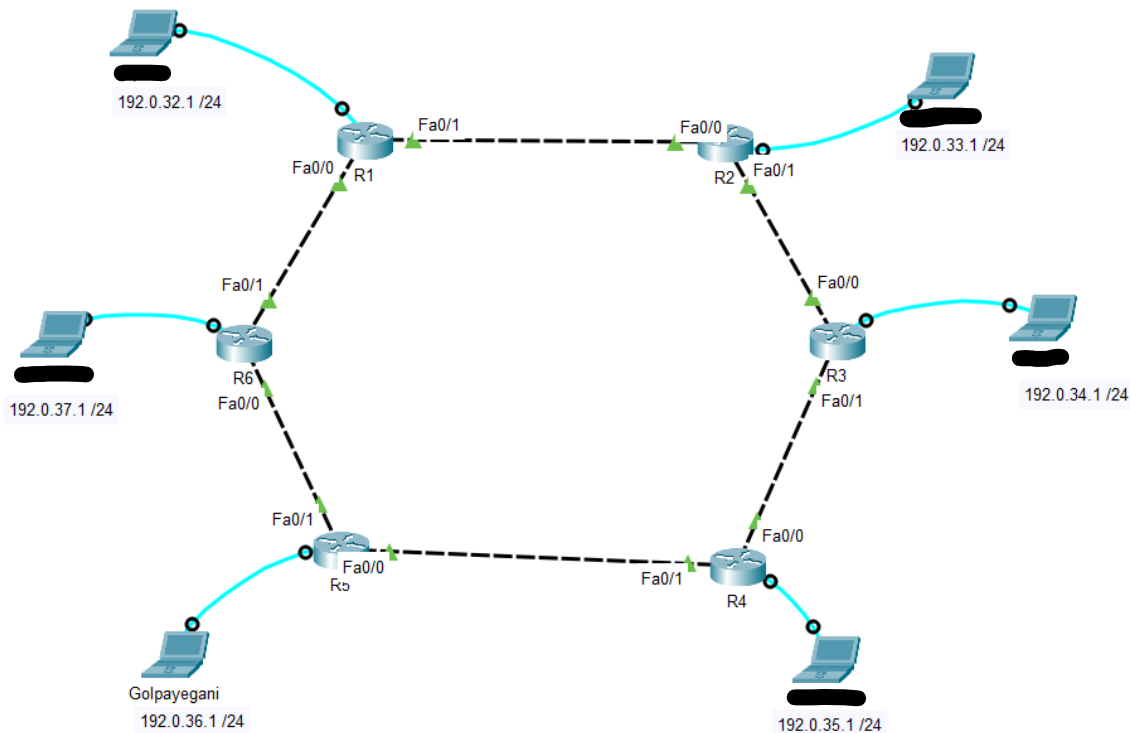
Gruppe (n.3)

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- Golpayegani Hossein
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Ziel

Verbinden Sie 6 Router, indem Sie das Protokoll RIP (Router Information Protokoll Version 2) implementieren, sodass alle Router alle anderen Router im Netzwerk anpingen können. Aus Sicherheitsgründen werden auf allen Routern Passwörter gesetzt.

Mapp



PC-Verbindung via Konsole, PING gemacht via Router to Router (gibt nur 2 Interface)



IP-Tabelle

Ich berechne auch die Netzwerk-IDs, die ich zur Implementierung des RIP-Protokolls benötige.

GERÄT	INTERFACE	IP ADRESSE	SUBNET	NETZWERK-ID
R1	Loop 0	192.0.32.1	255.255.255.0	192.0.32.0
	Fa 0/0	203.0.113.22	255.255.255.252	203.0.113.20
	Fa 0/1	203.0.113.1	255.255.255.252	203.0.113.0
R2	Loop 0	192.0.33.1	255.255.255.0	192.0.33.0
	Fa 0/0	203.0.113.2	255.255.255.252	203.0.113.0
	Fa 0/1	203.0.113.5	255.255.255.252	203.0.113.4
R3	Loop 0	192.0.34.1	255.255.255.0	192.0.34.0
	Fa 0/0	203.0.113.6	255.255.255.252	203.0.113.4
	Fa 0/1	203.0.113.9	255.255.255.252	203.0.113.8
R4	Loop 0	192.0.35.1	255.255.255.0	192.0.35.0
	Fa 0/0	203.0.113.10	255.255.255.252	203.0.113.8
	Fa 0/1	203.0.113.13	255.255.255.252	203.0.113.12
R5	Loop 0	192.0.36.1	255.255.255.0	192.0.36.0
	Fa 0/0	203.0.113.14	255.255.255.252	203.0.113.12
	Fa 0/1	203.0.113.17	255.255.255.252	203.0.113.16
R6	Loop 0	192.0.37.1	255.255.255.0	192.0.37.0
	Fa 0/0	203.0.113.21	255.255.255.252	203.0.113.20
	Fa 0/1	203.0.113.18	255.255.255.252	203.0.113.16



Konsole-Verbindung zum Router.

Um zu sehen, mit welchem Port der PC mit dem Router verbunden ist, öffne ich unter Windows den Geräte-Manager und prüfe den Eintrag Anschlüsse (COM & LPT) bzw. den Namen des USB-zu-Seriell-Ports.

Einstellungen die Hostname „R1“

The screenshot shows the Windows Device Manager window with the 'Ports (COM & LPT)' section expanded, highlighting 'USB Serial Port (COM12)'. Overlaid on this is a PuTTY terminal window connected to the router. The terminal output shows the router's boot sequence, including the IOS version (12.4(24)T) and the configuration commands being entered: `enable`, `configure terminal`, `hostname R1`, and `login`. A third window, 'PuTTY Configuration', is also visible, showing the 'Serial' connection type selected for COM12 at 9600 baud.

```
ROUTER(config)#enable password
ROUTER(config)#line console 0
ROUTER(config-line)#password
ROUTER(config-line)#login
```

Aktivation lookback 0

```
R1(config-if)#interface loopback 0
R1(config-if)#ip add
R1(config-if)#ip address 192.0.32.1 255.255.255.0
R1(config-if)#
R1#
```

```
R3(config)#int loopback 0
R3(config-if)#ip
*Jan 1 00:58:51.611: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
changed state to upadd
% Incomplete command.

R3(config-if)#ip add 192.0.34.1 255.255.255.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
```

```
R4(config)#int loopback 0
R4(config-if)#ip address 192.0.35.1 255.255.255.0
R4(config-if)#exit
R4(config)#exit
R4#
```



```
COM6 - PuTTY
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int loopback 0
Router(config-if)#
*Jan 1 00:41:44.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
changed state to up
Router(config-if)#ip address 192.0.37.1 255.255.255.0
Router(config-if)#exit
Router(config)#exit
Router#
*Jan 1 00:44:37.243: %SYS-5-CONFIG_I: Configured from console by console
Router#ip interface brief
^
% Invalid input detected at '^' marker.

Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0          unassigned      YES unset    administratively down down
FastEthernet0/1          unassigned      YES unset    administratively down down
Serial0/0/0              unassigned      YES unset    administratively down down
Serial0/1/0              unassigned      YES unset    administratively down down
Serial0/2/0              unassigned      YES unset    administratively down down
Serial0/2/1              unassigned      YES unset    administratively down down
Loopback0                192.0.37.1      YES manual    up          up
```



Interface

Konfiguration Interface und der Kommentar (dann ich wisse was die Interface Verbringung ist)

```
R1(config)#interface fastEthernet 0/1
R1(config-if)#ip ad
R1(config-if)#ipadd
R1(config-if)#ip ad
R1(config-if)#ip add
R1(config-if)#ip address 203.0.113.1 255.255.255.252
R1(config-if)#no sh
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Jan  1 00:40:55.607: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
R1(config)#in
R1(config)#interface f
R1(config)#interface fastEthernet 0/0
R1(config-if)#ip add
R1(config-if)#ip address 203.0.113.22 255.255.255.252
R1(config-if)#no h
R1(config-if)#no sh
R1(config-if)#no shutdown
```

```
R1(config)#interface f
R1(config)#interface fastEthernet 0/1
R1(config-if)#de
R1(config-if)#desc
R1(config-if)#description Schwarz
R1(config-if)#exit
R1(config)#in
R1(config)#interface fa
R1(config)#interface fastEthernet 0/0
R1(config-if)#desc
R1(config-if)#description Komolka
R1(config-if)#
```

```
R3(config)#int fa0/0
R3(config-if)#ip add 203.0.113.6 255.255.255.252
R3(config-if)#no sh
R3(config-if)#exit
*Jan  1 00:57:03.659: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
exit
R3(config)#int fa0/1
R3(config-if)#203.0.113.9 255.255.255.252
^
% Invalid input detected at '^' marker.
R3(config-if)#ip add 203.0.113.9 255.255.255.252
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
*Jan  1 00:57:45.747: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
R3(config)#
```



[REDACTED]

```
R4(config)#interface fa0/1
R4(config-if)#ip address 203.0.113.13 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Jan  1 01:01:42.315: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Jan  1 01:01:43.315: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R4(config)#interface fa0/0
R4(config-if)#ip address 203.0.113.10 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit+
^
% Invalid input detected at '^' marker.

R4(config-if)#exit
R4(config)#
```

Golpayegani

```
Router>
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface loopback 0
Router(config-if)#
*Jan  1 00:42:12.671: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
Router(config-if)#ip address 192.0.36.1 255.255.255.0
Router(config-if)#interface Fa0/0
Router(config-if)#ip address 203.0.113.14 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
*Jan  1 00:53:52.583: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
Router(config-if)#interface Fa0/1
Router(config-if)# ip address 203.0.113.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#
*Jan  1 00:55:59.847: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
Router(config-if)#interface Fa0/1
Router(config-if)#no ip address
Router(config-if)#ip address 203.0.113.17 255.255.255.252
Router(config-if)#no shutdown
```



```
R6(config)#interface fa0/1
R6(config-if)#ip address 203.0.113.18 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#!
*Jan  1 01:00:58.951: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Jan  1 01:00:59.951: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R6(config-if)#interface fa0/0
R6(config-if)#ip address 2023.0.113.21 255.255.255.252
^
% Invalid input detected at '^' marker.

R6(config-if)#ip address 203.0.113.21 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#!
R6(config-if)#
*Jan  1 01:02:49.651: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
```



Konfiguration RIP

Konfiguration die Remote Information Protokoll Version 2 (nicht Version 1 weil nicht die Support Netzmaske hat. Die erste Version wurde 1988 erstellt, wobei IPv4-Adressen klassenbasiert waren und Subnetzmasken (SM) nicht implementiert waren.)

Ich gebe außerdem an, mit welchem Netzwerk mein Router bereits verbunden ist.

```
R1(config)#router rip
R1(config-router)#ver
R1(config-router)#version 2
R1(config-router)#net
R1(config-router)#network 203.0.113.0
R1(config-router)#net
R1(config-router)#network 203.0.113.20
R1(config-router)#
```

Und ich warte die RIP suchen die andren Router . Es ist nicht nötigen das in befehlen „Network“ sage auch di Subnetzmasken, weil es von Interface bekommen.

```
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#no auto-summary
R3(config-router)#network 203.0.113.4
R3(config-router)#network 203.0.113.8
R3(config-router)#
*Jan  1 01:02:22.475: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R3(config-router)#end
R3#
*Jan  1 01:02:27.895: %SYS-5-CONFIG_I: Configured from console by console
```

```
R4#conf term
Enter configuration commands, one per line
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 203.0.113.8
R4(config-router)#network 203.0.113.12
R4(config-router)#end
R4#
```


**Golpayegani**

```
Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.
*Jan  1 01:01:43.143: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
*Jan  1 01:02:49.783: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
% Incomplete command.

Router(config-router)#network 192.0.36.0
Router(config-router)#network 203.0.113.0
Router(config-router)#passive-interface Loopback0
Router(config-router)#end
```

COM6 - PuTTY

```
o up
*Jan  1 01:02:50.651: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R6(config-if)#interface loopback 0
R6(config-if)#ip address 192.0.37.1 255.255.255.0
R6(config-if)#
R6(config-if)#
R6(config-if)#router rip
R6(config-router)#version 2
R6(config-router)#no auto-summary
R6(config-router)#network 203.0.113.16
R6(config-router)#network 203.0.113.20
R6(config-router)#end
R6#
*Jan  1 01:17:59.675: %SYS-5-CONFIG_I: Configured from console by console
R6#write memory
Building configuration...

[OK]
R6#show ip int brief
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/0          203.0.113.21    YES manual    up            up
FastEthernet0/1          203.0.113.18    YES manual    up            up
Serial0/0/0               unassigned      YES unset      administratively down down
Serial0/1/0               unassigned      YES unset      administratively down down
Serial0/2/0               unassigned      YES unset      administratively down down
Serial0/2/1               unassigned      YES unset      administratively down down
Loopback0                 192.0.37.1      YES manual    up            up
R6#ping 192.0.32.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.32.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R6#ping 192.0.33.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.33.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```



DEBUG

Zu erste benutze ich die „**show ip router**“ um zu prüfen, ob mein Router Informationen von den anderen Routern erhalten hat.

```
Success rate is 0 percent (0/5)
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    203.0.113.0/30 is subnetted, 6 subnets
R       203.0.113.16 [120/4] via 203.0.113.2, 00:00:05, FastEthernet0/1
C       203.0.113.20 is directly connected, FastEthernet0/0
R       203.0.113.8 [120/2] via 203.0.113.2, 00:00:05, FastEthernet0/1
R       203.0.113.12 [120/3] via 203.0.113.2, 00:00:05, FastEthernet0/1
C       203.0.113.0 is directly connected, FastEthernet0/1
R       203.0.113.4 [120/1] via 203.0.113.2, 00:00:05, FastEthernet0/1
R       192.0.36.0/24 [120/4] via 203.0.113.2, 00:00:05, FastEthernet0/1
C       192.0.32.0/24 is directly connected, Loopback0
R       192.0.33.0/24 [120/1] via 203.0.113.2, 00:00:06, FastEthernet0/1
R1#
```

```
R2#show route rip
route-map rip not found
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    203.0.113.0/30 is subnetted, 6 subnets
R       203.0.113.16 [120/3] via 203.0.113.6, 00:00:04, FastEthernet0/1
R       203.0.113.20 [120/1] via 203.0.113.1, 00:00:23, FastEthernet0/0
R       203.0.113.8 [120/1] via 203.0.113.6, 00:00:04, FastEthernet0/1
R       203.0.113.12 [120/2] via 203.0.113.6, 00:00:04, FastEthernet0/1
C       203.0.113.0 is directly connected, FastEthernet0/0
C       203.0.113.4 is directly connected, FastEthernet0/1
R       192.0.36.0/24 [120/3] via 203.0.113.6, 00:00:04, FastEthernet0/1
C       192.0.33.0/24 is directly connected, Loopback0
R2#
```



```
R3#ping 203.0.113.9

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.9, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R3#
```

Golpayegani

```
R5#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    203.0.113.0/30 is subnetted, 5 subnets
R       203.0.113.20 [120/4] via 203.0.113.13, 00:00:08, FastEthernet0/0
R       203.0.113.8 [120/1] via 203.0.113.13, 00:00:08, FastEthernet0/0
C       203.0.113.12 is directly connected, FastEthernet0/0
R       203.0.113.0 [120/3] via 203.0.113.13, 00:00:08, FastEthernet0/0
R       203.0.113.4 [120/2] via 203.0.113.13, 00:00:08, FastEthernet0/0
C      192.0.36.0/24 is directly connected, Loopback0
R      192.0.33.0/24 [120/3] via 203.0.113.13, 00:00:08, FastEthernet0/0
R5#
```

```
COM6 - PuTTY
R6#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#router rip
R6(config-router)#version 2
R6(config-router)#network 203.0.113.16
R6(config-router)#network 203.0.113.20
R6(config-router)#ex
R6(config-router)#exit
R6(config)#exit
R6#show ip route
*Jan  1 01:30:48.203: %SYS-5-CONFIG I: Configured from console by nterface fa0/1
FastEthernet0/1 is up, line protocol is up
  Hardware is MV96340 Ethernet, address is 0021.d80c.5261 (bia 0021.d80c.5261)
  Internet address is 203.0.113.18/30
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, 100BaseTX/FX
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:13, output 00:00:07, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    109 packets input, 18702 bytes
      Received 109 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog
    0 input packets with dribble condition detected
  420 packets output, 34720 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
```

**Und PING mit andrem Router (R2-6)**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R1#ping 203.0.113.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.5, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R1#ping 203.0.113.6

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.6, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R1#ping 203.0.113.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R1#ping 203.0.113.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.5, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R1#ping 203.0.113.6

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.6, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R1#ping 203.0.113.9

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.9, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R1#ping 203.0.113.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R1#ping 203.0.113.13
```



R2#ping 203.0.113.22

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.22, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms

R2#ping 203.0.113.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms

R2#ping 203.0.113.6

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.6, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms

R2#ping 203.0.113.9

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.9, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms

R2#ping 203.0.113.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms

R2#ping 203.0.113.13

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.13, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms

R2#

R2#ping 203.0.113.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms

R2#ping 203.0.113.13

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.13, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms

R2#ping 203.0.113.21

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.21, timeout is 2 seconds:

.....

Success rate is 0 percent (0/5)

R2#ping 203.0.113.14

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.14, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms



```
R2#ping 203.0.113.21
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 203.0.113.21, timeout is 2 seconds:  
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

```
R2#ping 203.0.113.18
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 203.0.113.18, timeout is 2 seconds:  
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

```
R2#
```



```
R3#ping 203.0.113.6
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 203.0.113.6, timeout is 2 seconds:  
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

```
R3#
```

```
R3#ping 203.0.113.2
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 203.0.113.2, timeout is 2 seconds:  
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
```

```
R3#
```

```
R3#ping 203.0.113.2
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 203.0.113.2, timeout is 2 seconds:  
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
```

```
R3#
```

```
R3#ping 203.0.113.2
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 203.0.113.2, timeout is 2 seconds:  
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
```

```
R3#
```

```
R3#ping 203.0.113.9
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 203.0.113.9, timeout is 2 seconds:  
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
```

```
R3#
```

```
R3#ping 203.0.113.9
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 203.0.113.9, timeout is 2 seconds:  
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
```

```
R3#
```



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```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R5#ping 203.0.113.18

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.18, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R5#
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R5#ping 203.0.113.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R5#
```

```
R6#ping 192.0.36.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.36.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R6#ping 191.0.32.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 191.0.32.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R6#192.0.32.1
Trying 192.0.32.1 ...
% Destination unreachable; gateway or host down

R6#ping 203.0.113.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R6#ping 203.0.113.22

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.22, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R6#
```



Für Sicherheit mache ich auch „**tracerroute**“

```
Type escape sequence to abort.
Tracing the route to 203.0.113.5

  1 203.0.113.2 0 msec 0 msec *
R1#tracerroute 203.0.113.09

Type escape sequence to abort.
Tracing the route to 203.0.113.9

  1 203.0.113.2 0 msec 0 msec 0 msec
  2 203.0.113.6 4 msec 0 msec *
R1#tracerroute 203.0.113.13

Type escape sequence to abort.
Tracing the route to 203.0.113.13

  1 203.0.113.21 0 msec 0 msec 0 msec
  2 203.0.113.17 4 msec 0 msec 0 msec
  3 203.0.113.13 4 msec 0 msec *
R1#tracerroute 203.0.113.14

Type escape sequence to abort.
Tracing the route to 203.0.113.14

  1 203.0.113.21 0 msec 0 msec 0 msec
  2 203.0.113.17 4 msec 0 msec *
R1#tracerroute 203.0.113.21

Type escape sequence to abort.
Tracing the route to 203.0.113.21

  1 203.0.113.21 0 msec 0 msec *
R1#tracerroute 203.0.113.18

Type escape sequence to abort.
Tracing the route to 203.0.113.18

  1 203.0.113.21 0 msec 0 msec *
R1#tracerroute 203.0.113.10

Type escape sequence to abort.
Tracing the route to 203.0.113.10

  1 203.0.113.2 0 msec 0 msec 0 msec
  2 203.0.113.6 4 msec 0 msec 0 msec
  3 203.0.113.10 4 msec 0 msec *
R1#
```




R2#Traceroute 203.0.113.22

Type escape sequence to abort.
Tracing the route to 203.0.113.22

1 203.0.113.1 0 msec 0 msec *
R2#

R2#traceroute 203.0.113.14

Type escape sequence to abort.
Tracing the route to 203.0.113.14

1 203.0.113.6 0 msec 0 msec 0 msec
2 203.0.113.10 4 msec 0 msec 0 msec
3 203.0.113.14 4 msec 0 msec *
R2#

R2#traceroute 203.0.113.13

Type escape sequence to abort.
Tracing the route to 203.0.113.13

1 203.0.113.6 0 msec 0 msec 0 msec
2 203.0.113.10 4 msec 0 msec *
R2#

R2#traceroute 203.0.113.18

Type escape sequence to abort.
Tracing the route to 203.0.113.18

1 203.0.113.1 0 msec 0 msec 0 msec
2 203.0.113.21 4 msec 0 msec *
R2#



```
R3#traceroute 203.0.113.13
Type escape sequence to abort.
Tracing the route to 203.0.113.13
  1 203.0.113.10 0 msec * 0 msec
R3#
```

```
R3#traceroute 203.0.113.14
Type escape sequence to abort.
Tracing the route to 203.0.113.14
  1 203.0.113.10 0 msec 0 msec 4 msec
  2 203.0.113.14 0 msec * 0 msec
R3#
```

```
R3#traceroute 203.0.113.17
Type escape sequence to abort.
Tracing the route to 203.0.113.17
  1 203.0.113.10 4 msec 0 msec 0 msec
  2 203.0.113.14 4 msec * 0 msec
R3#
```

```
R3#traceroute 203.0.113.18
Type escape sequence to abort.
Tracing the route to 203.0.113.18
  1 203.0.113.10 0 msec 0 msec 4 msec
  2 203.0.113.14 0 msec 0 msec 4 msec
  3 203.0.113.18 0 msec * 0 msec
R3#
```

```
R3#traceroute 203.0.113.21
Type escape sequence to abort.
Tracing the route to 203.0.113.21
  1 203.0.113.5 0 msec 0 msec 4 msec
  2 203.0.113.1 0 msec 0 msec 4 msec
  3 203.0.113.21 0 msec * 0 msec
R3#
```

```
R3#traceroute 203.0.113.22
Type escape sequence to abort.
Tracing the route to 203.0.113.22
  1 203.0.113.5 0 msec 0 msec 4 msec
  2 203.0.113.1 0 msec * 0 msec
R3#
```



```
R4#traceroute 203.0.113.22

Type escape sequence to abort.
Tracing the route to 203.0.113.22

  1 203.0.113.14 0 msec 0 msec 4 msec
  2 203.0.113.18 0 msec 0 msec 0 msec
  3 203.0.113.22 4 msec 0 msec *
R4#traceroute 203.0.113.6

Type escape sequence to abort.
Tracing the route to 203.0.113.6

  1 203.0.113.9 0 msec 0 msec *
R4#traceroute 203.0.113.21

Type escape sequence to abort.
Tracing the route to 203.0.113.21

  1 203.0.113.14 0 msec 0 msec 0 msec
  2 203.0.113.18 4 msec 0 msec *
R4#
```

```
R4#traceroute 203.0.113.22

Type escape sequence to abort.
Tracing the route to 203.0.113.22

  1 203.0.113.14 0 msec 0 msec 4 msec
  2 203.0.113.18 0 msec 0 msec 0 msec
  3 203.0.113.22 4 msec 0 msec *
R4#traceroute 203.0.113.6

Type escape sequence to abort.
Tracing the route to 203.0.113.6

  1 203.0.113.9 0 msec 0 msec *
R4#traceroute 203.0.113.21

Type escape sequence to abort.
Tracing the route to 203.0.113.21

  1 203.0.113.14 0 msec 0 msec 0 msec
  2 203.0.113.18 4 msec 0 msec *
R4#
```



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```
R5#
R5#
R5#
R5# traceroute 203.0.113.21

Type escape sequence to abort.
Tracing the route to 203.0.113.21

  1 203.0.113.18 4 msec * 0 msec
R5# traceroute 203.0.113.10

Type escape sequence to abort.
Tracing the route to 203.0.113.10

  1 203.0.113.13 0 msec * 0 msec
R5# traceroute 203.0.113.6

Type escape sequence to abort.
Tracing the route to 203.0.113.6

  1 203.0.113.13 4 msec 0 msec 0 msec
  2 203.0.113.9 4 msec * 0 msec
R5# traceroute 203.0.113.2

Type escape sequence to abort.
Tracing the route to 203.0.113.2

  1 203.0.113.18 0 msec 0 msec 4 msec
  2 203.0.113.22 0 msec 0 msec 0 msec
  3 203.0.113.2 0 msec * 0 msec
R5# traceroute 203.0.113.22

Type escape sequence to abort.
Tracing the route to 203.0.113.22

  1 203.0.113.18 0 msec 0 msec 4 msec
  2 203.0.113.22 0 msec * 0 msec
R5#
```



COM6 - PuTTY

```
R6#traceroute 203.0.113.22
Type escape sequence to abort.
Tracing the route to 203.0.113.22
  1 203.0.113.22 0 msec * 0 msec
R6#traceroute 203.0.113.1
Type escape sequence to abort.
Tracing the route to 203.0.113.1
  1 203.0.113.22 0 msec * 0 msec
R6#traceroute 203.0.113.2
Type escape sequence to abort.
Tracing the route to 203.0.113.2
  1 203.0.113.22 4 msec 0 msec 0 msec
  2 203.0.113.2 4 msec * 0 msec
R6#traceroute 203.0.113.5
Type escape sequence to abort.
Tracing the route to 203.0.113.5
  1 203.0.113.22 0 msec 0 msec 4 msec
  2 203.0.113.2 0 msec * 0 msec
R6#traceroute 203.0.113.6
Type escape sequence to abort.
Tracing the route to 203.0.113.6
  1 203.0.113.22 0 msec 0 msec 0 msec
  2 203.0.113.2 0 msec 4 msec 0 msec
  3 203.0.113.6 0 msec * 0 msec
R6#traceroute 203.0.113.9
Type escape sequence to abort.
Tracing the route to 203.0.113.9
  1 203.0.113.17 0 msec 4 msec 0 msec
  2 203.0.113.13 0 msec 4 msec 0 msec
  3 203.0.113.9 0 msec * 0 msec
```



Sicherheit

Konfiguration der Sicherheitsrichtlinien des Routers mit Krypto des Password.

```
R1#en
R1#enable
R1#conf
R1#configure t
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#lin
R1(config)#line v
R1(config)#line vty 0
R1(config)#line vty 0 15
R1(config-line)#pass
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#
```

```
R1#en
R1#enable
R1#conf
R1#configure t
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#lin
R1(config)#line v
R1(config)#line vty 0
R1(config)#line vty 0 15
R1(config-line)#pass
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#line
R1(config)#line con
R1(config)#line console 0
R1(config-line)#pas
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#serv
R1(config)#service pa
R1(config)#service pass
R1(config)#service password-encryption
R1(config)#
```



```
Router(config)#hostname R4
R4(config)#int loopback 0
R4(config-if)#
*Jan  1 00:48:42.483: %LINEPROTO-5-UPDOWN: Line
changed state to up
R4(config-if)#service-password encryption
      ^
% Invalid input detected at '^' marker.

R4(config-if)#service password-encryption
R4(config)#service password-encryption
R4(config)#enable secret cisco
R4(config)#username alan privilege 1 password ci
R4(config)#enable secret class
R4(config)#line console 0
R4(config-line)#login localk
      ^
% Invalid input detected at '^' marker.

R4(config-line)#login local
R4(config-line)#exit
R4(config)#line vty 0 4
R4(config-line)#login local
R4(config-line)#transport input telnet
R4(config-line)#exit
R4(config)#exit
R4#
*Jan  1 00:55:08.735: %SYS-5-CONFIG_I: Configure
R4#copy run start
Destination filename [startup-config]?
Building configuration...

[OK]
```

```
1485 bytes copied in 0.724 secs (2051 bytes/sec)
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#hostname R3
R3(config)#service password-encryption
R3(config)#username jonas privilege 1 password cisco
R3(config)#enable secret class
R3(config)#exit
R3#copy
*Jan  1 00:55:29.479: %SYS-5-CONFIG_I: Configured from console by cons
% Incomplete command.

R3#copy run start
Destination filename [startup-config]?
Building configuration...

[OK]
R3#
```




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COM4 - PuTTY

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status        Protoco
1
FastEthernet0/0          203.0.113.14    YES manual up            up
FastEthernet0/1          203.0.113.17    YES manual up            up
Serial0/0/0              unassigned      YES unset  administratively down down
Serial0/1/0              unassigned      YES unset  administratively down down
Serial0/2/0              unassigned      YES unset  administratively down down
Serial0/2/1              unassigned      YES unset  administratively down down
Loopback0                192.0.36.1      YES manual up            up

Router#ping 203.0.113.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
Router#ping 203.0.113.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R5
R5(config)#no ip domain-lookup
R5(config)#service password-encryption
R5(config)#banner motd ^CUnauthorized access prohibited!^C
R5(config)#enable secret cisco36
R5(config)#line console 0
R5(config-line)# password cisco36
R5(config-line)# login
R5(config-line)# exec-timeout 10 0
R5(config-line)#line vty 0 4
R5(config-line)# password cisco36
R5(config-line)# login
R5(config-line)# transport input telnet
R5(config-line)#end
R5#
*Jan  1 01:32:56.603: %SYS-5-CONFIG_I: Configured from console by console
R5#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...

[OK]
R5#
```



```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R6
R6(config)#conf t
^
% Invalid input detected at '^' marker.

R6(config)#no ip domain-lookup
R6(config)#enable secret ciscoen
R6(config)#service password-encryption
R6(config)#!
R6(config)#line con 0
R6(config-line)#password cisco
R6(config-line)#login
R6(config-line)#exit
R6(config)#line vty 0 4
R6(config-line)#password cisco
R6(config-line)#login
R6(config-line)#exit
R6(config)#exit
R6#
*Jan  1 00:55:48.939: %SYS-5-CONFIG_I: Configured from console by console
R6#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
```